

Analysis I

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Lecture 1 – 22/01/26

Analysis is about things that are small, things that are large, and quantify how small or large they are.

§1 Numerical sequences and series

§1.1 Basics

Definition 1.1. A sequence $(x_n)_{n \in \mathbb{N}}$ on a set X is an enumerated list where each element is in X .

In this chapter, $X \subseteq \mathbb{R}$ or $X \subseteq \mathbb{C}$.

What does it mean for a sequence (x_n) to converge to a point $x \in X$? What does it mean for (x_n) to diverge to infinity?

For convergence, we want $|x_n - x|$ to be small. But what is small? In fact, we want $|x_n - x|$ to be arbitrarily small, i.e. smaller than any $\varepsilon > 0$ we choose. What really matters is the *tail* of the sequence – we can always ignore the first N terms, and we should pick N depending on ε .

For divergence to infinity, we want $|x_n|$ to get arbitrarily large, i.e. larger than any $L > 0$ we choose. Again, we ignore the first N terms depending on L .

Definition 1.2. We say (x_n) converges to some finite x , $x_n \rightarrow x$, if, for any $\varepsilon > 0$, there exists $N(\varepsilon)$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$.

We say (x_n) diverges to infinity if, for any $L > 0$, there exists $N(L)$ such that $x_n > L$ for all $n \geq N$, analogously for negative infinity. For complex sequences, we must consider $|x_n|$.

- Example 1.3**
- Take $x_n = \frac{1}{n}$, then $x_n \rightarrow 0$ because $|x_n - 0| = \frac{1}{n} < \varepsilon$ for any $n \geq N = 1 + \lceil \frac{1}{\varepsilon} \rceil$.
 - Take $x_n = \frac{1}{2^n}$, then $x_n \rightarrow 0$ because $|x_n - 0| = \frac{1}{2^n} < \varepsilon$ for any $n \geq N = \max(1 + \lceil \log_2(\frac{1}{\varepsilon}) \rceil, 1)$.
 - Take $x_n = in$, then $x_n \rightarrow \infty$, since $\frac{x_n}{i} = n > L$ for any $n \geq N = 1 + \lceil L \rceil$.
 - Take $x_n = (-1)^n$, then x_n doesn't converge or diverge to infinity.

Lemma 1.4

If it exists, the limit of (x_n) is unique.

Proof. Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$. Take $\varepsilon > 0$, then we have

$$\begin{aligned} \exists N_1 = N_1(\varepsilon) \text{ s.t. } |x_n - a| < \varepsilon \quad \forall n \geq N_1 \\ \exists N_2 = N_2(\varepsilon) \text{ s.t. } |x_n - b| < \varepsilon \quad \forall n \geq N_2 \end{aligned}$$

Now take $N = \max(N_1, N_2)$. We have

$$\begin{aligned} |a - b| &= |a - x_n + x_n - b| \\ &\leq |x_n - a| + |x_n - b| \\ &< 2\varepsilon \end{aligned}$$

for all $\varepsilon > 0$. It follows that $|a - b| = 0$, so $a = b$. ■

This justifies the notation $\lim_{n \rightarrow \infty} x_n = x$.

Proposition 1.5 (Sandwich theorems)

Let $(x_n), (y_n), (z_n)$ be real sequences. Then

- If $x_n \leq y$ for all n and $x_n \rightarrow x$, then $x \leq y$. Note that $x_n < y$ does not imply $x < y$.
- If $x_n \rightarrow x$, $z_n \rightarrow x$ and $x_n \leq y_n \leq z_n$ for all n , then $y_n \rightarrow x$.

Proof. Left as an exercise. ■

Lemma 1.6

Let (x_n) be a complex sequence. Then $x_n \rightarrow x$ if and only if $\operatorname{Re}(x_n) \rightarrow \operatorname{Re}(x)$ and $\operatorname{Im}(x_n) \rightarrow \operatorname{Im}(x)$.

Proof. Recall that, for $z \in \mathbb{C}$, $|z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$.

$\Rightarrow$ Use $|\operatorname{Re}(z)| \leq |z|$ and $|\operatorname{Im}(z)| \leq |z|$ with $z = x_n - x$.

$\Leftarrow$ Use $|z| \leq |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$, again with $z = x_n - x$. Fixing $\varepsilon > 0$, we can get $|x_n - x| < 2\varepsilon$, so we're done. ■

Lemma 1.7

Let $x_n \rightarrow x$, $y_n \rightarrow y$. Then $x_n + y_n \rightarrow x + y$, $x_n y_n \rightarrow xy$.

Proof. The first part is analogous to the above result on complex sequences and is an exercise.

For the second part, we have

$$\begin{aligned} |x_n y_n - xy| &= |x_n y_n - x y_n + x y_n - xy| \\ &\leq |y_n| |x_n - x| + |x| |y_n - y| \end{aligned}$$

Now, for $\varepsilon > 0$, we can pick an N to make $|x_n - x| < \varepsilon$, $|y_n - y| < \varepsilon$. Hence

$$\begin{aligned} |x_n y_n - xy| &< \varepsilon(|x| + |y_n|) \\ &< \varepsilon(|x| + Y) \quad \text{by lemma 1.9, } Y > 0 \end{aligned}$$

and so picking $\varepsilon = \frac{\varepsilon'}{|x|+Y}$ shows the result. ■

Definition 1.8 (Bounded sequences). We say (x_n) is *bounded* if there exists $M > 0$ such that $|x_n| \leq M$ for all n , or, equivalently, if $\sup |x_n| \leq M$.

Lemma 1.9

Convergent sequences are bounded.

Proof. Take $\varepsilon = 1$, then there exists N such that $|x_n - x| \leq 1$ for any $n \geq N$, and so $|x_n| \leq 1 + |x|$ for any $n \geq N$. But for $n < N$, $|x_n| \leq \max_{1 \leq i < N} |x_i| < \infty$. ■

Definition 1.10 (Monotone sequences). We say a real sequence (x_n) is *monotone* if either

- it is increasing;
- or it is decreasing.

Proposition 1.11 (Monotone convergence theorem)

Every bounded monotone sequence converges.

Proof sketch. Without loss of generality, suppose the sequence is strictly increasing. Using the least upper bound property, we can show that $x_n \rightarrow \sup\{x_n\}$. ■

Can we strengthen to non-monotonic sequences? Clearly not fully, since $x_n = (-1)^n$ is bounded but not convergent. But, if we delete all the even terms, then it indeed becomes convergent.

§1.2 Bolzano-Weierstrass theorem

Theorem 1.12 (Bolzano-Weierstrass)

If (x_n) is a real bounded sequence, then it has at least one convergent subsequence.

Definition 1.13 (Subsequences). Given a sequence (x_n) , a subsequence of (x_n) is a sequence of the form $(x_{n_k})_{k \in \mathbb{N}}$, where (n_k) is a strictly increasing sequence of natural numbers.

Lemma 1.14

If $x_n \rightarrow x$, then any subsequence (x_{n_k}) of (x_n) must converge to x .

Proof. Since $n_k < n_{k+1}$, $n_{k+1} \geq n_k + 1$. By induction, we can show $n_k \geq k$.

Now, take $\varepsilon > 0$, so we have $N(\varepsilon)$ such that $|x_n - x| < \varepsilon$ for any $n \geq N$. So, if $k \geq N$, then $n_k \geq k \geq N$, and so $|x_{n_k} - x| < \varepsilon$, so $\lim_{k \rightarrow \infty} x_{n_k} = x$ as required. ■

Proposition 1.15 (Nested interval property)

Let $(I_n)_{n \in \mathbb{N}}$ be a sequence of closed intervals in $\mathbb{R}$ satisfying $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, and suppose $I_n = [a_n, b_n]$. If $b_n - a_n = |I_n| \rightarrow 0$ as $n \rightarrow \infty$, then $\bigcap_{n \in \mathbb{N}} I_n$ contains exactly one point.

Proof. We can see that (a_n) and (b_n) are bounded monotone sequences, so we probably want to apply [proposition 1.11](#). Indeed, we have $a_n \rightarrow a$, $b_n \rightarrow b$. Since limits preserve inequalities and $a_n \leq b_n$, we have $a \leq b$. Now we are ready to show the result.

- *Existence.* For all $k \geq n$, $a_k \in [a_k, b_k] \subseteq [a_n, b_n]$, and so $a_n \leq a_k \leq b_n$. Taking k to infinity, we have $a_n \leq a \leq b_n$ for every n , so $a \in I_n$ for every n , so it is in their intersection.
- *Uniqueness.* We know $b_n - a_n \rightarrow b - a$. But we know $b_n - a_n \rightarrow 0$, so $a = b$. Hence, if x is in the intersection, $a_n \leq x \leq b_n$ for all n , so $a \leq x \leq a$, and hence $x = a$ as required. ■

Proof of theorem. We are given (x_n) and $M > 0$ such that $|x_n| \leq M$ for any n . We will construct a series of nested intervals from which we can sample our subsequence and apply [proposition 1.15](#).

Let $a_1 = -M$, $b_1 = M$, so $I_1 = [-M, M] \supset (x_n)$. Now take $c = \frac{a_1 + b_1}{2}$. Then, at least one of the halves $[a_1, c]$ and $[c, b_1]$ will have infinitely many elements of (x_n) – we take I_2 to be a half that has infinitely many elements. Continue inductively to get (I_n) , which are nested by construction and all of which have infinitely many elements of the sequence – and by [proposition 1.15](#), there exists exactly one x in their intersection.

We choose (x_{n_k}) as follows: pick n_1 such that $x_{n_1} \in I_1$. Then I_2 has infinitely many elements of (x_n) with index greater than n_1 , so pick $n_2 > n_1$ such that $x_{n_2} \in I_2$. Continue inductively.

We have $x_{n_k} \in I_k$ for every k , so $x_{n_k} \in \bigcap_{n \leq k} I_n$, and so $x_{n_k} \rightarrow x$. ■

Corollary 1.16

The Bolzano-Weierstrass theorem holds in $\mathbb{C}$.

§1.3 Cauchy sequences

A Cauchy sequence is a sequence in which the elements are getting closer as the sequence progresses.

Definition 1.17. A sequence (x_n) on $X \subseteq \mathbb{C}$ is *Cauchy* if, for any $\varepsilon > 0$, there exists N such that, for all $n, m \geq N$, $|x_n - x_m| < \varepsilon$.

Example • Take $x_n = \frac{1}{n}$. Assume wlog $m \geq n$. Then

$$\left| \frac{1}{n} - \frac{1}{m} \right| = \frac{1}{n} - \frac{1}{m} = \frac{m - n}{mn} \leq \frac{1}{n} < \varepsilon$$

and this holds for any $m \geq n \geq N = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$. So (x_n) is Cauchy.

- Take $x_n = (-1)^n$. This is not a Cauchy sequence – take $n = 2k$, $m = 2k + 1$ for any $k \in \mathbb{N}$, so $|x_n - x_m| = 2$. Hence we can't take $\varepsilon < 2$.
- Take (x_n) on $\mathbb{Q}$ defined by truncation of the decimal expansion of $\sqrt{2}$. Assume wlog $m \geq n$, so

$$|x_n - x_m| < 10^{-(n-1)} \rightarrow 0$$

Note that this sequence does not converge in $\mathbb{Q}$.

Lemma 1.18

If (x_n) is Cauchy, then it is bounded.

Proof. Take $\varepsilon = 1$. Then we have N such that

$$|x_n - x_N| < 1$$

for any $n \geq N$ (we have fixed $m = N$). So $|x_n| < 1 + |x_N|$. ■

Lemma 1.19

A complex sequence (x_n) is Cauchy if and only if $(\operatorname{Re} x_n)$ and $(\operatorname{Im} x_n)$ are both Cauchy.

Lemma 1.20

If $x_n \rightarrow x$, then (x_n) is Cauchy.

Proof. For any $\varepsilon > 0$, we have N such that

$$\begin{aligned} |x_n - x_m| &\leq |x_n - x| + |x_m - x| \\ &\leq 2\varepsilon \end{aligned}$$

for $n, m \geq N$. This completes the proof. $\blacksquare$

Is the converse true? Not in $\mathbb{Q}$ at least – see $\sqrt{2}$ example. But we can still aspire to the converse being true in $\mathbb{R}$.

Theorem 1.21 (Completeness of $\mathbb{R}$ and $\mathbb{C}$)

Every Cauchy sequence in $\mathbb{R}$ or $\mathbb{C}$ converges.

Proof. Suppose (x_n) is Cauchy. Then it is bounded, and so by Bolzano-Weierstrass it has a convergent subsequence (x_{n_k}) converging to some x . We claim $x_n \rightarrow x$. Observe

$$|x_n - x| \leq |x_n - x_{n_k}| + |x_{n_k} - x|$$

Taking $\varepsilon > 0$, Cauchy gives $|x_n - x_{n_k}| < \varepsilon$ for $n, n_k \geq N_1$, and $x_{n_k} \rightarrow x$ gives $|x_{n_k} - x| < \varepsilon$ for $k \geq N_2$. Fixing $k \geq N_2$ such that $n_k \geq N_1$, we have $|x_n - x| < 2\varepsilon$ for every $n \geq N_1$, which suffices for the result. $\blacksquare$

Remark. This is nice – it's our first way to prove convergence without having to guess the limit.

§1.4 Series and convergent series

Definition 1.22. Let (a_n) be a sequence in $\mathbb{R}$ or $\mathbb{C}$. We say that $\sum_{n=1}^{\infty} a_n$ is a *series*, and we say it converges if the sequence of partial sums $s_k = \sum_{n=1}^k a_n$ converges to some finite s . In this case, s is called the sum of the series, and we write

$$s = \sum_{n=1}^{\infty} a_n$$

Example • $\sum_{n=1}^{\infty} n$ does not converge: $s_k = \frac{n(n+1)}{2}$.

- geometric series: $\sum_{n=1}^{\infty} r^n$ converges iff $|r| < 1$. We have $s_k = \frac{1-r^{k+1}}{1-r}$ (unless $r = 1$).
- $\sum_{n=1}^{\infty} n \text{ fty } \frac{1}{n(n+1)}$ converges to 1. We have

$$s_k = \sum_{n=1}^k \frac{1}{n(n+1)} = \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{k+1} \rightarrow 1$$

Lemma 1.23

Fix $\lambda \in \mathbb{C}$. If $\sum a_n$ and $\sum b_n$ converge, then $\sum(\lambda a_n + b_n)$ also converges

Proof. Exercise. ■

As usual, we should understand that the tail of the series is what is significant to its convergence.

Lemma 1.24

If $a_n = b_n$ for all $n \geq N$, then $\sum a_n$ converges if and only if $\sum b_n$ converges.

Proof. We have

$$\begin{aligned} r_k &= \sum_{n=1}^k b_n \\ &= \sum_{n=1}^{N-1} b_n + \sum_{n=N}^k a_n \\ &= \underbrace{\sum_{n=1}^{N-1} a_n}_{s_k} + \sum_{n=N}^k (b_n - a_n) \end{aligned}$$

Taking limits, we conclude that

$$\lim_{k \rightarrow \infty} r_k = \lim_{k \rightarrow \infty} s_k + C$$

which concludes. ■

Proposition 1.25 (n^{th} term test)

If $\sum a_n$ converges, then $a_n \rightarrow 0$.

Remark 1.26. The converse does not hold. Consider the harmonic series. We have

$$s_k = \sum_{n=1}^k \frac{1}{n}$$

Observe

$$\begin{aligned} s_{2k} &= s_k + \frac{1}{n+1} + \cdots + \frac{1}{n+n} \\ &\geq s_k + \frac{1}{2n} + \cdots + \frac{1}{2n} \\ &= s_k + \frac{1}{2} \end{aligned}$$

Hence s_k is not Cauchy, so it does not converge.

Proof. $\sum a_n$ converges $\implies s_k$ converges, so (s_k) is Cauchy. Then $s_{k+1} - s_k = a_{k+1} \rightarrow 0$. ■

Let's first focus on tests for the convergence of $\sum a_n$, where $a_n \geq 0$ for all n .

Proposition 1.27 (Comparison test)

Suppose we have sequences satisfying $0 \leq b_n \leq a_n$ for all n . Then $\sum a_n$ converges $\implies \sum b_n$ converges.

Proof. Set s_k and r_k to be the partial sums of a_n, b_n respectively. Since $a_n, b_n \geq 0$, (s_k) and (r_k) are increasing. Furthermore, $s_k \rightarrow s \implies s_k \leq s$. But

$$b_n \leq a_n \implies \sum_{n=1}^k b_n \leq \sum_{n=1}^k a_n$$

so $r_k \leq s_k$, and hence r_k converges by the monotone convergence theorem. ■

Example 1.28

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. We have

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=1}^{\infty} \frac{1}{(n+1)^2}$$

But then

$$0 \leq \frac{1}{(n+1)^2} \leq \frac{1}{n(n+1)}$$

and we know $\sum \frac{1}{n(n+1)} \rightarrow 1$. So $\frac{1}{n^2}$ converges by the comparison test.

Proposition 1.29 (Root test)

Suppose $a_n \geq 0$ for all n . Suppose there exists a such that $a = \lim_{n \rightarrow \infty} a_n^{1/n}$. If

- $a < 1$, then $\sum a_n$ converges;
- $a > 1$, then $\sum a_n$ diverges;
- $a = 1$, then the test is inconclusive.

Proof. If $a > 1$, then we have N such that $a_n^{1/n} > 1$ for any $n \geq N$. But then $a_n > 1$, so $a_n \not\rightarrow 0$, and we fail the n^{th} term test. Hence the series diverges.

If $a < 1$, then we can find $r \in \mathbb{R}$ with $a < r < 1$. Then we have N such that $a_n^{1/n} < r$ for any $n \geq N$. But then $a_n < r^n$ for all $n \geq N$. By the comparison test on the tails with a geometric series, $\sum a_n$ converges. ■

- Example 1.30**
- Consider $\sum \frac{1}{2^n}$, then $a_n^{1/n} = \frac{1}{2}$, so $\sum a_n$ converges.
 - Consider $\sum 4^n$, then $a_n^{1/n} = 4$, so $\sum a_n$ does not converge.

Proposition 1.31 (Ratio test)

If $a_n \geq 0$ for all n and $\frac{a_{n+1}}{a_n} \rightarrow a$ for some finite a , then if

- $a < 1$, then $\sum a_n$ converges;
- $a > 1$, then $\sum a_n$ diverges;
- $a = 1$, then the test is inconclusive.

Example 1.32 • If we check $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$, we see both lie in the inconclusive regions for both tests.

- Let us consider $\sum \frac{n}{2^n}$. The ratio test gives

$$\frac{a_{n+1}}{a_n} = \frac{\frac{n+1}{2^{n+1}}}{\frac{n}{2^n}} = \frac{n+1}{2n} \rightarrow \frac{1}{2}$$

which is nice. If instead we use the root test, then

$$\left(\frac{n}{2^n}\right)^{\frac{1}{n}} = \frac{n^{1/n}}{2} \rightarrow \frac{1}{2}$$

But actually showing that $n^{1/n} \rightarrow 1$ is non-trivial... we need to apply l'Hôpital. Observe

$$n^{1/n} = e^{\log(n^{1/n})} = e^{\frac{1}{n} \log n} \rightarrow e^0 = 1$$

- If the ratio test is conclusive for a sequence, then the root test is also conclusive, but not the converse:

$$a_n = \begin{cases} 2^{-n} & n \text{ even} \\ 2^{-n+1} & n \text{ odd} \end{cases}$$

Then the limit of ratios does not exist, but the root test shows convergence.

Proposition 1.33 (Integral test)

Suppose $f : [1, \infty) \rightarrow [0, \infty)$ is a continuous decreasing function (and hence integrable in $[1, N]$ for each $N \in \mathbb{N}$), and define $a_n = f(n)$. Then

$$\sum a_n \text{ converges} \iff \lim_{n \rightarrow \infty} \int_1^n f(t) dt \text{ exists}$$

Furthermore, as $k \rightarrow \infty$,

$$\sum_{n=1}^k a_n - \int_1^k f(x) dx \rightarrow l \text{ for some } l \in [0, f(1)]$$

Proof. Observe (DIAGRAM)

$$\begin{aligned} s_k - a_1 &= \sum_{n=2}^k a_n \leq \int_1^k f(t) dt \\ s_{k-1} &= \sum_{n=1}^{k-1} a_n \geq \int_1^k f(t) dt \end{aligned}$$

Now

$\implies \int_1^k f(t) dt \leq s_{k-1} \leq s$ since s_k is increasing and converges by assumption. Hence $(\int_1^k f(t) dt)_k$ converges by MCT.

$\Leftarrow s_k \leq a_1 + \int_1^k f(t) dt \leq a_1 + \lim_{k \rightarrow \infty} \int_1^k f(t) dt$. Hence s_k converges by MCT.

For the last part, let

$$b_k = \sum_{n=1}^k a_n - \int_1^k f(x) dx$$

We have

$$b_k - b_{k-1} = a_k - \int_{k-1}^k f(t) dt = f(k) \cdot 1 - \int_{k-1}^k f(t) dt \leq 0$$

and

$$\begin{aligned} \sum_{n=1}^{k-1} a_n &\geq \int_1^k f(t) dt \\ \implies \sum_{n=1}^k a_n - \int_1^k f(t) dt &\geq a_k \\ \implies b_k &\geq a_k = f(k) \geq 0 \end{aligned}$$

Hence b_k is decreasing and bounded below, so converges to some $l \in \mathbb{R}$. Since $0 \leq b_k \leq b_1 = a_1 = f(1)$, we conclude $0 \leq l \leq f(1)$. $\blacksquare$

Example 1.34 • $\sum \frac{1}{n^p}$ converges if and only if $p > 1$, since

$$\lim_{x \rightarrow \infty} \int_1^x \frac{1}{t^p} dt = \lim_{x \rightarrow \infty} \begin{cases} \frac{1}{1-p} x^{1-p} & p \neq 1 \\ \log x & p = 1 \end{cases}$$

• $\sum \frac{1}{n \log n}$ diverges, since

$$\int \frac{1}{t \log t} dt = \log \log t + C$$

• $\sum \frac{1}{n \log^2 n}$ converges, since

$$\int \frac{1}{t \log^2 t} dt = -\frac{1}{\log t} + C$$

Proposition 1.35 (Cauchy condensation test, non-examinable)

Let $a_n \geq 0$ and (a_n) be decreasing. Then $\sum a_n$ converges if and only if $\sum 2^n a_{2^n}$ converges.

Proof. Observe

$$\int_1^x f(t) dt = \log 2 \int_0^{2^x} 2^t f(2^t) dt$$

Then, by the integral test, we conclude

$$\sum a_n \text{ converges} \iff \lim_{y \rightarrow \infty} \int_0^y 2^t f(2^t) dt \text{ exists}$$

While we may be tempted to try and conclude here, it is not known that $g(t) = 2^t f(2^t)$ is decreasing.

MIUST CHECK BACK. ■

Proposition 1.36 (Alternating series test)

Let (a_n) be a decreasing sequence, and suppose $a_n \geq 0$, $a_n \rightarrow 0$. Then $\sum (-1)^{n+1} a_n$ converges.

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Proof. Set $s_k = \sum^k (-1)^{n+1} a_n$. Then

$$s_{2k} = \underbrace{(a_1 - a_2)}_{\geq 0} + \underbrace{(a_3 - a_4)}_{\geq 0} + \cdots + \underbrace{(a_{2k-1} - a_{2k})}_{\geq 0}$$

and so $s_2 \leq s_4 \leq \dots$. Likewise, $s_1 \geq s_3 \geq \dots$. Moreover, $s_{2k+1} - s_{2k} = a_{2k+1} \geq 0$, so that

$$s_2 \leq s_{2k} \leq s_{2k+1} \leq s_1$$

so s_{2k+1} is bounded below by s_2 , and s_{2k} is bounded above by s_1 . Hence, by the monotone convergence theorem, we have $s_{2k} \rightarrow s$, $s_{2k+1} \rightarrow \tilde{s}$. Now, note $s_{2k+1} - s_{2k} = a_{2k+1} \rightarrow 0$, and so $s = \tilde{s}$.

Finally, let us show s_n converges. We have for every $\varepsilon > 0$ an N_1 such that $|s_{2k} - s| < \varepsilon$ for any $k \geq N_1$, and an N_2 such that $|s_{2k+1} - s| < \varepsilon$ for any $k \geq N_2$. But then, taking $N = 2 \max(N_1, N_2) + 1$, we can see that $|s_k - s| < \varepsilon$ for any $k \geq N$. ■

Proposition 1.37 (Dirichlet test)

Let (a_n) be a decreasing sequence, and suppose $a_n \geq 0$, $a_n \rightarrow 0$. Let (b_n) be another sequence such that $s_k = (\sum^n b_n)$ is bounded. Then $\sum a_n b_n$ converges.

Proof. An exercise. ■

Definition 1.38 (Absolute convergence). We say $\sum a_n$ is *absolutely convergent* if $\sum |a_n|$ is convergent.

Lemma 1.39

If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

Proof. Let $s_k = \sum^n a_n$, $r_k = \sum^n |a_n|$. Observe

$$\begin{aligned} |s_k - s_l| &= \left| \sum_{n=l+1}^k a_n \right| \\ &\leq \sum_{n=l+1}^k |a_n| \\ &= r_k - r_l \\ &= |r_k - r_l| \end{aligned}$$

Hence r_k convergent $\implies r_k$ Cauchy $\implies s_k$ Cauchy $\implies s_k$ convergent. ■

Remark. This is awesome, because we can show absolute convergence using all our tests for series with positive terms (the converse is clearly false).

We sometimes call sequences that are convergent but not absolutely convergent conditionally convergent. Conditionally convergent sequences are weird.

Example 1.40

Consider

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

Rearranging terms on the right, we produce

$$\begin{aligned} & \left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \left(\frac{1}{7} - \frac{1}{14}\right) + \dots \\ &= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \frac{1}{14} \\ &= \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots\right) \end{aligned}$$

so we've produced a different value by rearranging the terms. For a conditionally convergent series, the order of the terms matters.

Proposition 1.41 (Rearrangements under absolute convergence)

Let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. Let $a'_n = a_{\sigma(n)}$. Then, if $\sum a_n$ is absolutely convergent, $\sum a'_n = \sum a_n$.

Proof. Let $s_k = \sum_{n=1}^k a_n$, and suppose $s_k \rightarrow s$. Then, for any ε we have N such that, for $k \geq N$,

$$|s_k - s| < \varepsilon \text{ and } \sum_{k>N} |a_n| < \varepsilon$$

Furthermore, there exists $N_1 \geq N$ such that $a_1, \dots, a_N$ are included in $a'_1, \dots, a'_{N_1}$. Hence, for $m \geq N_1$,

$$\begin{aligned} \sum_{n=1}^m a'_n &= \underbrace{\sum_{n=1}^N a_n}_{s_N} + (m - N \text{ terms of } (a_n) \text{ with index } \geq N) \\ \Rightarrow \left| \sum_{n=1}^m a'_n - s \right| &\leq \underbrace{|s_N - s|}_{< \varepsilon} + \underbrace{\sum_{k>N} |a_n|}_{< \varepsilon} \\ &< 2\varepsilon \end{aligned}$$

which was what we wanted. ■

§2 Continuity of functions

§2.1 Limits of functions

Take $f : \mathbb{C} \rightarrow \mathbb{C}$. We want to define what we mean by $f(z) \rightarrow y$ as $z \rightarrow a$, even if $a \notin X$. A classic example is $\frac{\sin x}{x}$ at $a = 0$ – even though 0 is not in the domain, there are points in the domain that are very close to 0, so it feels like we should be able to assign 0 some value.

Definition 2.1 (Accumulation point). Let $X \subseteq \mathbb{C}$. We say X is an *accumulation point* for X if, for any $\delta > 0$, there exists $z \in X \setminus \{a\}$ such that $|z - a| < \delta$. Otherwise, we say a is *isolated*.

Example 2.2

- Take $X = \{0\} \cup [1, 2]$ – then 0 is an isolated point, but all points in $[1, 2]$ are accumulation points.
- Take $S = \{|z| = 1\}$ – these are accumulation points for the set $D = \{|z| < 1\}$ (also, every point in D is an accumulation point for D).

Lemma 2.3

Let $X \subseteq \mathbb{C}$, $a \in \mathbb{C}$. Then a is an accumulation point for X if and only if there exists (x_n) in $X \setminus \{a\}$ such that $x_n \rightarrow a$.

Let's return to thinking about $f(z) \rightarrow y$ as $z \rightarrow a$. We should be able to make sense of this if a is either in the domain of f , or is an accumulation point for the domain of f .

Definition 2.4. Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Let $a \in X$ or a an accumulation point of X . Then we say $f(z) \rightarrow y$ as $z \rightarrow a$ if, for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon)$ such that $|z - a| < \delta, z \in X \implies |f(z) - y| < \varepsilon$.

Furthermore, if a is an accumulation point but not in the domain, we say $f(z) \rightarrow \infty$ as $z \rightarrow a$ if

- $f(z)/|f(z)| \rightarrow l$ for some $l \in \mathbb{C}$;
- and, for any $L > 0$, there exists $\delta = \delta(L)$ such that $|z - a| < \delta, z \in X \implies |f(z)| > L$.

Remark 2.5. If a is an isolated point of X , then we can always find δ sufficiently small such that $|z - a| < \delta, z \in X \iff z = a$. Hence $f(z) \rightarrow f(a)$ as $z \rightarrow a$ trivially.

If f diverges at a , then $a \notin X$, which is why we require $a \in X$ as part of the definition.

Example 2.6

Take $f(x) = \frac{\sin x}{x}$ with domain $\mathbb{R} \setminus \{0\}$. We claim $\frac{\sin x}{x} \rightarrow 1$ as $x \rightarrow 0$.

Recall the geometric argument that shows $\cos x < \frac{\sin x}{x} < 1$ for every $x \in (0, \frac{\pi}{2})$. Hence

$$\left| \frac{\sin x}{x} - 1 \right| < |1 - \cos x| = 2 \sin^2 \left(\frac{x}{2} \right) < \frac{x^2}{2}$$

Hence, for any $\varepsilon > 0$, choosing $\delta = \sqrt{2\varepsilon}$ gives the result.

Lemma 2.7

Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$, and take $a \in \mathbb{C}$ either in X or an accumulation point of X . Then

- $f(z) \rightarrow y$ as $z \rightarrow a$ if and only if $f(z_n) \rightarrow y$ for every sequence (z_n) on X with $z_n \rightarrow a$;
- $f(z)$ diverges as $z \rightarrow a$ if and only if $f(z_n)$ diverges for every sequence (z_n) on X with $z_n \rightarrow a$.

Lemma 2.8

If $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ has a limit at $a \in \mathbb{C}$, then this limit is unique.

Proof. Suppose $f(z) \rightarrow x$ and $f(z) \rightarrow y$ as $z \rightarrow a$. Then

$$|x - y| \leq |x - f(z)| + |f(z) - y|$$

Taking the limit as $z \rightarrow a$, we have

$$|x - y| \leq 0 + 0$$

so $x = y$ as required. ■

Remark. This justifies the notation $\lim_{z \rightarrow a} f(z) = y$.

Lemma 2.9

Let $f, g : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$, and $a \in \mathbb{C}$ either in X or an accumulation point of X . Suppose $\lim_{z \rightarrow a} f(z) = y$, $\lim_{z \rightarrow a} g(z) = x$. Then

$$\begin{aligned} \lim_{z \rightarrow a} f(z) + g(z) &= y + x \\ \lim_{z \rightarrow a} f(z)g(z) &= yx \end{aligned}$$

If $g(z) \neq 0$ for any $z \in X$, and $x \neq 0$, then

$$\lim_{z \rightarrow a} \frac{f(z)}{g(z)} = \frac{y}{x}$$

§2.2 Continuity of functions

Definition 2.10. Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Then f is continuous at $a \in X$ if

$$f(a) = \lim_{z \rightarrow a} f(z)$$

or, equivalently, for every $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon)$ such that $|z - a| < \delta$, $z \in X \implies |f(z) - f(a)| < \varepsilon$.

Definition 2.11. Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. We say f is continuous if it is continuous at every point in X .

Example 2.12 • $f(z) = z$ is continuous.

• Consider

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

This function is not continuous.

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Lemma 2.13

Let $f, g : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ be continuous at $a \in \mathbb{C}$. Then, so are $\lambda f + g$ for $\lambda \in \mathbb{C}$ fixed, fg and, if $f(z) \neq 0$ for any $z \in X$, $1/f(z)$.

Lemma 2.14

Let $U, V \subseteq \mathbb{C}$, and $f : U \rightarrow V, g : V \rightarrow \mathbb{C}$. Suppose f is continuous at $a \in U$ and g is continuous at $f(a) \in V$ – then, $g \circ f : U \rightarrow \mathbb{C}$ is continuous at a .

Proof. Since g is continuous at $f(a)$, we have $\sigma = \sigma(\varepsilon)$ such that $y \in V, |y - f(a)| < \sigma$ implies $|g(y) - g(f(a))| < \varepsilon$. Furthermore, since f is continuous at a , we have for any σ a $\delta = \delta(\sigma) = \delta(\varepsilon)$ such that

$$z \in V, |z - a| < \delta \implies |f(z) - f(a)| < \sigma \implies |g(f(z)) - g(f(a))| < \varepsilon$$

■

There is another, equivalent notion of continuity which is extremely useful.

Definition 2.15 (Sequential continuity). Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. We say f is *sequentially continuous* at $a \in X$ if $f(z_n) \rightarrow f(a)$ for all (z_n) on X with $z_n \rightarrow a$.

Proposition 2.16

Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Then f is continuous if and only if it is sequentially continuous.

Proof. $\implies$ By continuity of f , for any $\varepsilon > 0$ we have δ such that $|z - a| < \delta, z \in X$ implies $|f(z) - f(a)| < \varepsilon$. Take (z_n) on X with $z_n \rightarrow a$, then

■

§2.3 Extreme value theorem

Definition 2.17 (Closed sets). A set $X \subseteq \mathbb{C}$ is *closed* if all sequences (x_n) in X which converge in $\mathbb{C}$ have limits in X .

Example • The closed interval $[0, 1]$ is closed, while the open interval $(0, 1)$ is not.

- $[0, 1)$ and $(0, 1]$ are not closed.
- $[0, \infty)$ is closed, while $(0, \infty]$ is not.

Definition 2.18 (Bounded sets). We say $X \subseteq \mathbb{C}$ is *bounded* if there exists $M > 0$ such that $X \subseteq \{z \in \mathbb{C} : |z| \leq M\}$, or in other words $\sup_{z \in X} |z| \leq M$.

Example

$[0, 1]$, $(0, 1)$, $[0, 1)$ and $(0, 1]$ are all bounded, while $[0, \infty)$ and $(0, \infty)$ are not.

Proposition 2.19

Let $X \subseteq \mathbb{C}$ be a closed bounded set. If $f : X \rightarrow \mathbb{C}$ is continuous, then $f(X) \subseteq \mathbb{C}$ is a closed bounded subset of $\mathbb{C}$.

Proof. • $f(X)$ is bounded. Suppose not. Then, for each $n \in \mathbb{N}$, we can find $x_n \in X$ such that $|f(x_n)| > n$. Now, (x_n) is a sequence in X , so it must be bounded, so Bolzano-Weierstrass applies – there exists a convergent subsequence (x_{n_k}) such that $x_{n_k} \rightarrow x$ and, by closedness of X , $x \in X$. On the other hand, $|f(x_{n_k})| > n_k \geq k$, so $f(x_{n_k})$ cannot converge as $k \rightarrow \infty$. But then f is not continuous, since it is not sequentially continuous.

- $f(X)$ is closed. Take (y_n) in $f(X)$, and suppose it converges to some $y \in \mathbb{C}$ – we want to show $y \in f(X)$. Since $y_n \in f(X)$, we have $x_n \in X$ such that $f(x_n) = y_n$, and (x_n) has a subsequence (x_{n_k}) converging to $x \in X$ by the closure of X and the Bolzano-Weierstrass theorem. By continuity of f ,

$$y = \lim_{k \rightarrow \infty} y_{n_k} = \lim_{k \rightarrow \infty} f(x_{n_k}) = f\left(\lim_{k \rightarrow \infty} x_{n_k}\right) = f(x)$$

which was what we wanted. ■

Theorem 2.20 (Extreme value theorem)

Let $X \subseteq \mathbb{R}$ be a closed bounded set, and let $f : X \rightarrow \mathbb{R}$ be continuous. Then there exist $a, b \in X$ satisfying

$$f(a) = \sup f(X); \quad f(b) = \inf f(X)$$

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Proof. We will consider specifically the supremum, since the infimum is analogous.

Let $M = \sup f(X)$. By definition, $M - \frac{1}{n}$ is not an upper bound on $f(X)$ for any $n \in \mathbb{N}$, that is, we can find a sequence (y_n) with elements in $f(X)$ such that $M - \frac{1}{n} < y_n \leq M$, which produces a sequence (x_n) on X such that $M - \frac{1}{n} < f(x_n) \leq M$. Now take limits and use the fact that f is continuous – we have $M \leq \lim_{n \rightarrow \infty} f(x_n) \leq M$.

Finally, by [proposition 2.19](#), we know that, since (y_n) is a sequence in $f(X)$, its limit is also in the image of X . ■

Remark. We know that the sup and inf exist – the key point is that f actually achieves both of these.

§2.4 Intermediate value theorem

Theorem 2.21 (Intermediate value theorem)

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous then, for any y satisfying $f(a) \leq y \leq f(b)$, there exists x satisfying $a \leq x \leq b$ such that $f(x) = y$.

Proof. If f is constant, then the result is clear. If not, wlog let $f(a) \leq f(b)$. Set $S = \{x \in [a, b] : f(x) \leq y\}$. Then $a \in S$, so S is non-empty – furthermore, S is bounded above by b , so we have $c = \sup S \in [a, b]$. We claim that $f(c) = y$. Observe:

- We claim $f(c) \leq y$. Suppose not. Then $\varepsilon = f(c) - y > 0$. By continuity of f , we have δ such that $|x - c| < \delta$, $x \in [a, b]$ implies $|f(x) - f(c)| < \varepsilon$, so $f(x) > f(c) - \varepsilon = y$, so $x \notin S$. This means $(c - \delta, c) \cap S = \emptyset$, which contradicts the definition of c (in particular that c is the least upper bound).
- We claim $f(c) \geq y$. Suppose not. Then $\varepsilon = y - f(c) > 0$. By continuity of f , we have δ such that $|x - c| < \delta$, $x \in [a, b]$ implies $|f(x) - f(c)| < \varepsilon$, so $f(x) < f(c) + \varepsilon = y$. But then $f(c + \frac{\delta}{2}) < y$, so $c \leq c + \frac{\delta}{2} \in S$, which contradicts the definition of c (in particular that c is an upper bound).

The result follows. ■

Remark. A nice interpretation of this theorem is that continuous functions from reals to reals map intervals to intervals.

Example

We can use this result to show the existence of n^{th} roots. Take $t > 0$ and $N \in \mathbb{N}$, and consider

$$\begin{aligned} f : [0, 1+t] &\rightarrow \mathbb{R} \\ x &\mapsto x^N \end{aligned}$$

This is continuous, and $f(0) = 0 < t < (1+t)^N = f(1+t)$. Hence, there exists a c satisfying $c^N = t$ by the intermediate value theorem. Note that we do not have uniqueness.

Definition 2.22 (Monotone functions). A function $f : [a, b] \rightarrow \mathbb{R}$ is *monotone* if it is either increasing, $x_1 \leq x_2 \implies f(x_1) \leq f(x_2)$; or decreasing, $x_1 \leq x_2 \implies f(x_2) \leq f(x_1)$.

Proposition 2.23 (Inverse function theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and strictly monotone. Let $c = \min(f(a), f(b))$, $d = \max(f(a), f(b))$. Then $f : [a, b] \rightarrow [c, d]$ is a bijection, and $f^{-1} : [c, d] \rightarrow [a, b]$ is continuous and strictly monotone.